# Laws of Exponents

From Basic to Advanced

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# Session Flow: Master the Exponents

A 50-minute student-centered sequence based on Buckingham's PGC Framework.

**Lesson Roadmap (50 min)**

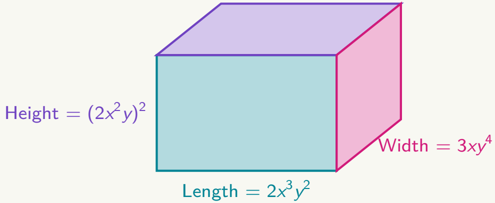
- **05 min – Phase 1:** Novelty & Curiosity (Geometric Volume Puzzle)
- **10 min – Phase 2:** Expectation & Modeling (Rules 1 to 11)
- **20 min – Phase 3:** Effort & Scaffolding (Easy $\rightarrow$ Very Hard Challenges)
- **15 min – Phase 4:** Recognition & Feedback (Error Analysis & Peer Review)

# 1. Novelty & Curiosity | Why Do We Need Laws of Exponents?

**The Geometric Complexity Problem**

Suppose a 3D rectangular box has the following algebraic dimensions:

- **Length:** $2x^3y^2$
- **Width:** $3xy^4$
- **Height:** $(2x^2y)^2$



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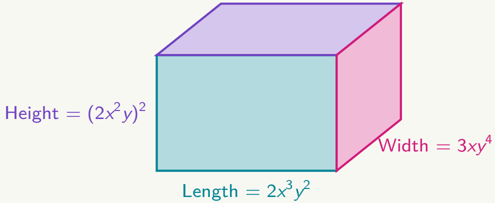


To calculate the Volume
$V =$ Length $\times$ Width $\times$ Height:

$$V = (2x^3y^2) \cdot (3xy^4) \cdot (2x^2y)^2$$

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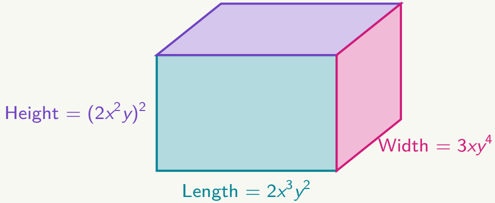


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## Peer Discussion (2 min)

Without writing out endless strings of variables, how can we simplify this expression quickly and accurately?

### Rule Definitions

1. $a^0 = 1 \quad (a \neq 0)$
2. $a^1 = a$
3. $a^n \cdot a^k = a^{n+k}$

### Class Exercises (Unsolved)

**Example 1:** Simplify

$$(7x^4y^{-2})^0 = \underline{\hspace{3cm}}$$

**Example 2:** Simplify

$$(-15xy^3)^1 = \underline{\hspace{3cm}}$$

**Example 3:** Simplify

$$(3x^4) \cdot (5x^7) = \underline{\hspace{3cm}}$$

Think-Pair-Share: Why does any non-zero term to the power of 0 equal 1?

## Rule Definitions

4. $\dfrac{a^k}{a^m} = a^{k-m}$
5. $(a^k)^n = a^{k\cdot n}$
6. $a^{-n} = \dfrac{1}{a^n}$

## Class Exercises (Unsolved)

**Example 4:** Simplify

$$\frac{12x^9}{4x^3} = \underline{\hspace{3cm}}$$

**Example 5:** Simplify

$$(x^5)^4 = \underline{\hspace{3cm}}$$

**Example 6:** Evaluate

$$4^{-3} = \underline{\hspace{3cm}}$$

Notice how negative exponents represent repeated division or reciprocals.

## 2. Expectation & Discovery | Rules 7 to 9: Products & Fractions

### Rule Definitions

7. $\dfrac{1}{a^{-n}} = a^n$
8. $(a \cdot b)^n = a^n b^n$
9. $\left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$

### Class Exercises (Unsolved)

**Example 7:** Simplify

$$\frac{5}{x^{-4}} = \underline{\hspace{3cm}}$$

**Example 8:** Expand

$$(2x^3y^4)^3 = \underline{\hspace{3cm}}$$

**Example 9:** Expand

$$\left(\frac{3x^2}{y^5}\right)^2 = \underline{\hspace{3cm}}$$

### Rule Definitions

10. $$\frac{a \cdot c^{-n}}{b} = \frac{a}{b \cdot c^{n}}$$

11. $$\frac{a}{b \cdot c^{-n}} = \frac{a \cdot c^{n}}{b}$$

### Class Exercises (Unsolved)

**Example 10:** Rewrite with positive exponents:

$$\frac{8x^{-3}}{2y^{2}} = \underline{\qquad\qquad\qquad}$$

**Example 11:** Rewrite with positive exponents:

$$\frac{10x^{3}}{5y^{-2}} = \underline{\qquad\qquad\qquad}$$

Rule of thumb: Moving a factor across the fraction line flips the sign of its exponent!

# Master Reference Matrix: All 11 Rules

## Rules 1 – 6

- **R1:** $a^0 = 1$
- **R2:** $a^1 = a$
- **R3:** $a^n \cdot a^k = a^{n+k}$
- **R4:** $\dfrac{a^k}{a^m} = a^{k-m}$
- **R5:** $(a^k)^n = a^{k \cdot n}$
- **R6:** $a^{-n} = \dfrac{1}{a^n}$

## Rules 7 – 11

- **R7:** $\dfrac{1}{a^{-n}} = a^n$
- **R8:** $(a \cdot b)^n = a^n b^n$
- **R9:** $\left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$
- **R10:** $\dfrac{a \cdot c^{-n}}{b} = \dfrac{a}{b \cdot c^n}$
- **R11:** $\dfrac{a}{b \cdot c^{-n}} = \dfrac{a \cdot c^n}{b}$

# 3. Effort & Scaffolding | Level 1: Basic Application

**[Level: Easy – Individual Work | 4 min]**

Simplify each expression and identify which rule number (1–11) you used:

**(a)**

$$x^5 \cdot x^8$$

**(b)**

$$\frac{y^{10}}{y^4}$$

**(c)**

$$(z^3)^4$$

**Rules Reference Panel**

**R1:** $a^0 = 1$ **R2:** $a^1 = a$ **R3:** $a^n a^k = a^{n+k}$ **R4:** $\frac{a^k}{a^m} = a^{k-m}$ **R5:** $(a^k)^n = a^{kn}$ **R6:** $a^{-n} = \frac{1}{a^n}$

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# 3. Effort & Scaffolding | Level 2: Multi-Rule Combination

**[Level: Medium – Pair Work | 5 min]**

Simplify completely. In your notebook, list every rule number (1–11) used step-by-step:

$$\text{Simplify:} \qquad \frac{\left(2x^3y^2\right)^3}{4x^4y^8}$$

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**Hint for Pairs**

Apply Power of a Product (Rule 8) and Power of a Power (Rule 5) to the numerator first!

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# 3. Effort & Scaffolding | Level 3: Advanced Challenge

**[Level: Hard – Small Group | 6 min]**

Simplify and express the final answer using **only positive exponents**:

$$\text{Simplify:} \quad \left(\frac{3x^{-2}y^{4}}{6x^{3}y^{-1}}\right)^{-2}$$

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**Strategy A (Inside-Out)**

Simplify inside the parentheses first using quotient rules, then apply outer power.

**Strategy B (Outside-In)**

Flip the fraction to change $-2$ to $+2$ using Rule 9 & negative laws, then expand.

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# 3. Effort & Scaffolding | Level 4: Master Synthesis

**[Level: Very Hard – Master Challenge | 5 min]**

Solve the master problem and state all rules applied along your solution path:

$$\text{Simplify completely:} \qquad \frac{\left(4x^{-3}y^{5}\right)^{-2} \cdot \left(2x^{2}y^{-1}\right)^{4}}{\left(8x^{-4}y^{2}\right)^{-1}}$$

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**Goal**

Can you reduce this entire expression to a single simplified fraction with positive exponents?

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# 4. Recognition & Feedback | Error Analysis: "Spot the Mistake"

**Become the Examiner (5 min)**

Each line below contains a common algebraic mistake. Identify the error and write the correct process!

1. **Student A claims:**

$$x^3 \cdot x^4 = x^{12} \quad \textbf{(Incorrect!)}$$

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$$\frac{x^6}{x^{-2}} = x^4 \quad \textbf{(Incorrect!)}$$

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**Correction Keys:** (1) Add powers $\rightarrow x^7$. (2) Power applies to 2 $\rightarrow 8x^9$. (3) Subtract negative power $\rightarrow x^{6-(-2)} = x^8$.

# 4. Recognition & Feedback | Peer Assessment Protocol

**Notebook Swap (5 min):** Exchange notebooks with your partner and check their Level 2 & Level 3 solutions.

1. **Praise:** Point out one step where your peer clearly applied a rule correctly.
2. **Growth Point:** Identify any sign or power multiplication error.
3. **Verification:** Sign their page once reviewed.

**Session Summary**

- Exponent laws reduce complex algebraic models (like 3D volumes) to simple forms.
- Negative exponents shift terms across fraction lines.
- Always apply powers to numerical coefficients as well as variable bases!

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| Negative Power | Crossing the fraction line inverts exponent sign ($a^{-n} = \frac{1}{a^n}$). |
| Power of a Power | Multiply exponents together when raising a power ($(a^k)^n = a^{k \cdot n}$). |

**Next Class: Unlocking Radical Laws through Fractional Exponents!**

# 50 Minutes of Focused Individual Work

## Task Objectives & Instructions

- **Individual Focus:** Work independently on your assigned task/homework.
- **Self-Paced Progression:** Move sequentially through the problem sets or task guidelines.
- **Resource Management:** Consult your reference notes, class slides, or guidelines before asking questions.

## Expectations

- Silent and continuous effort.
- Complete all required steps in your notebook/file.

## Teacher Support

- Raise your hand for targeted 1-on-1 feedback.
- Check progress against the success criteria.

**Timer:** 50:00 min – Maximize your active working time!